SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 3 – Class Test

Course Code:	DSA0301	**Course Title:**	Natural Language Processing
CO Assessed:	CO2 – Manipulation	**Assessment:**	Assessment Tool 3 – Class Test
Weightage:	20% of CIA		
CO5 Statement:	Design inflectional, derivational, finite-state parsing, stemming algorithms and for analyse morphological methods. (BL-3)		
Student Name:	B.Jahnavi	**Reg. No.:**	192325043
Section / Batch:	4^{th} year /AIML	**Date:**	09/08/26

Code of Conduct

I B. Jahnavi / 192325043 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: B.Jahnavi

Instructions

- Read all questions carefully before attempting the answers.
- Answer all five questions. Each question carries equal marks.
- Show all intermediate steps, calculations, probability computations, tagging decisions, and justifications wherever applicable.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
N-gram Models and Word Counting for Language Modeling	N-gram Construction, Word Frequency Analysis, Probability Estimation	1
Unsmoothed N-grams and Smoothing Techniques in Syntax Analysis	Unsmoothed N-grams, Backoff, Deleted Interpolation	2
Entropy and Language Uncertainty in NLP Systems	Information Theory, Entropy Measurement, Language Prediction	3
English Word Classes and Tagsets for Linguistic Analysis	Word Categories, POS Categories, English Tagsets	4
Part-of-Speech Tagging Techniques and Applications	POS Tagging, Rule-Based Tagging, Stochastic Tagging, Transformation-Based Tagging	5

Annexure A – Class Test

1. A document processing system encounters the words “talked”, “talking”, and “talks”. Apply inflectional morphology rules and Porter Stemmer steps to derive the normalized root form. Identify the suffixes removed and explain the role of morphological normalization in improving information retrieval accuracy.

Objective

A document processing system can use inflectional morphology and Porter Stemming to reduce different grammatical forms of the same word to a common root. This improves information retrieval because documents containing different forms can be matched with the same search term.

Morphological Rules

talk + -ed → talked: past-tense inflection.
talk + -ing → talking: present participle/continuous form.
talk + -s → talks: third-person singular present.
These are inflectional suffixes because they change grammatical form without changing the basic lexical meaning.

Analysis

Input	Root	Suffix	Function	Type	Normalized stem
talked	talk	-ed	Past tense	Inflectional	talk
talking	talk	-ing	Present participle	Inflectional	talk
talks	talk	-s	3^{rd} person singular present	Inflectional	talk

Porter Stemmer Steps

1. talked
talked → talk
Remove -ed.
Intermediate form: talk
Final stem: talk

2. talking
talking → talk
Remove -ing.
Intermediate form: talk
Final stem: talk

3. talks
talks → talk
Remove -s.
Intermediate form: talk
Final stem: talk

Final Output

talked → talk + ed → talk
talking → talk + ing → talk
talks → talk + s → talk

All three words are normalized to talk.

Importance for Information Retrieval

If a user searches for talk, the normalized index can also retrieve documents containing talked, talking, and talks. This reduces vocabulary variation and improves retrieval recall and consistency.

2. Given the input words: “relational”, “conditional”, “happiness”, “running”, design a hybrid morphological processing pipeline using derivational rules, inflectional analysis, finite-state morphological parsing, and Porter Stemmer. Decompose each word into root and affixes, represent transitions using finite-state parsing logic, and apply stemming steps to obtain the final normalized forms. Analyze how each technique contributes to accurate root extraction.

Objective

A hybrid morphological pipeline combines derivational analysis, inflectional analysis, finite-state parsing, and Porter Stemming to obtain consistent normalized stems.

Processing Pipeline

Input Word
↓
Morphological Analysis
↓
Identify Root + Affixes
↓
Finite-State Validation
↓
Porter Stemmer
↓
Final Normalized Stem

Morphological Analysis

Word	Root	Affix	Type	Final porter stem
relational	relate	-ion /-al	Derivational	relate
conditional	condition	-al	Derivational	condit
happiness	happy	-ness	Derivational	happi
running	run	-ing	Inflectional	run

1. Relational
Morphological structure:
relate + -ion + -al → relational
Root/base: relate
-ion: derivational suffix forming a noun.
-al: derivational suffix forming an adjective.
Word class: adjective.
Porter processing:
relational → relate → relat
Final stem: relat

2. Conditional

Morphological structure:
condition + -al → conditional
Root/base: condition
-al: derivational suffix.
Word class: adjective.
Porter processing removes -al:
conditional → condition → condit
Final stem: condit

3. Happiness

Morphological structure:
happy + -ness → happiness
Root/base: happy
-ness: derivational suffix forming a noun.
Spelling adjustment occurs: happy → happi before -ness.
Porter processing:
happiness → happi
Final stem: happi

4. Running

Morphological structure:
run + -ing → running
Root: run
-ing: inflectional suffix.
Double consonant is handled during morphological normalization.
Porter processing:
running → run
Final stem: run

Finite-State Parsing

The finite-state parser can use the following general transitions:
S0 → Root
Root → Derivational Suffix
Root → Inflectional Suffix
Suffix → Accept
Accept → Porter Stemmer

For the individual words:
relational:
S0 → relate → -ion → -al → Accept → relat
conditional:
S0 → condition → -al → Accept → condit
happiness:
S0 → happy → -ness → Accept → happi
running:
S0 → run → -ing → Accept → run

Final Output

Input	Morphological type	Final stem
relational	Derivational	relate
conditional	Derivational	condit
happiness	Derivational	happi
running	Inflectional	run

Contribution of Each Technique

- ✓ Derivational analysis identifies suffixes such as -ion, -al, and -ness.
- ✓ Inflectional analysis identifies grammatical suffixes such as -ing.
- ✓ Finite-state parsing validates possible sequences of roots and affixes.
- ✓ Porter Stemmer removes remaining suffixes and produces compact indexing stems.

3. An enterprise-level search engine optimization (SEO) system processes web content containing variations such as “optimize”, “optimization”, “optimized”, and “optimizing”. The system must ensure consistent keyword indexing and ranking. Develop a morphology-driven solution integrating inflectional and derivational analysis, finite-state morphological parsing for rule validation, and Porter Stemmer for normalization. Apply the approach to the given inputs and determine how the system achieves improved search relevance and reduced redundancy.

Objective

An SEO system must index related keyword forms consistently. Morphological analysis identifies derivational and inflectional changes, while finite-state parsing validates the structures and Porter Stemming produces normalized stems.

Morphological Analysis

word	Root	Affix	Type	Final stem
optimize	optim	-ize	Derivational	optim
optimization	optimize	-ation	Derivational	optim
optimized	optimize	-ed	Inflectional	optim
optimizing	optimize	-ing	Inflectional	optim

1. Optimize
Morphological structure:
optim + -ize → optimize
-ize is a derivational suffix.
It forms a verb.
Porter processing removes the final -e:
optimize → optim
Final stem: optim

2. Optimization
Morphological structure:
optimize + -ation → optimization
-ation is a derivational suffix.
It forms a noun.
Porter stemming progressively removes the derivational ending:
optimization → optim
Final stem: optim

3. Optimized
Morphological structure:
optimize + -ed → optimized
-ed is an inflectional suffix representing past tense/past participle.
Porter processing:
optimized → optimize → optim
Final stem: optim

4. Optimizing
Morphological structure:
optimize + -ing → optimizing
-ing is an inflectional suffix.
Porter processing:
optimizing → optimize → optim
Final stem: optim

Finite-State Transitions

```
S0 → optim
    |
    ├─── -ize → optimize → Accept
    ├─── -ation → optimization → Accept
    ├─── -ed → optimized → Accept
    └─── -ing → optimizing → Accept
            ↓
       Porter Stemmer
            ↓
          Optim
```

Final Output
optimize → optim
optimization → optim
optimized → optim
optimizing → optim

Effect on Search Relevance

Normalizing all forms to optim reduces redundant index entries. A query involving optimize can therefore match documents containing optimization, optimized, or optimizing. This increases recall and provides more consistent keyword grouping.

4. A large-scale information retrieval system processes input words: “compute”, “computer”, “computing”, and “computation” to enhance search accuracy.
Design a hybrid morphological pipeline integrating inflectional and derivational analysis, finite-state morphological parsing, and Porter Stemmer.
Decompose each word into root and affixes, represent transformations using state transitions, and apply stemming rules for normalization.
Ensure consistent root extraction across all variations for efficient indexing.
Determine the morphological breakdown, parsing transitions, and final normalized forms for each word.

Objective

The information retrieval system should identify both derivational and inflectional variations and normalize related forms for efficient indexing.

Morphological Analysis

word	Root	Affix	Type	Word class	Final stem
compute	compute	---	Base	Verb	comput
computer	compute	-er	Derivation	Noun	comput
computing	compute	-ing	Inflectional	Verb	comput
computation	compute	-ation	Derivational	Noun	comput

1. Compute
compute
Base verb.
No additional derivational or inflectional suffix.
Porter removes the final -e:
compute → comput
Final stem: comput

2. Computer
compute + -er → computer
-er is a derivational suffix.
Forms a noun referring to an agent/device.
Porter processing:
computer → comput
Final stem: comput

3. Computing
compute + -ing → computing
-ing is an inflectional suffix.
Represents the present participle/continuous form.
Porter processing:
computing → comput
Final stem: comput

4. Computation
compute + -ation → computation
-ation is a derivational suffix.
Forms a noun.
Porter processing removes the derivational ending:
computation → comput
Final stem: comput

Finite-State Transitions

```
S0 → compute → Accept
       |
       ├── -er  → computer     → Accept
       ├── -ing → computing    → Accept
       └── -ation → computation → Accept
```

```
All accepted forms
       ↓
Porter Stemmer
       ↓
     Comput
```

Final Output
compute → comput
computer → comput
computing → comput
computation → comput

Interpretation

The system recognizes that -er and -ation are derivational, while -ing is inflectional. Despite these differences, Porter stemming maps all four forms to comput, allowing them to share an index entry.

5.A text mining system processes input words: “decide”, “decision”, “decisive”, and “deciding” to support semantic clustering and analysis.
Construct a morphology-based framework combining derivational rules, inflectional transformations, finite-state parsing, and Porter stemming.
Analyze how suffix variations affect meaning and grammatical structure while maintaining root consistency. Apply the approach to extract roots, validate transformations, and normalize the dataset. Derive the morphological structure, state transitions, and final stem for each input word.

Objective

A text mining system can combine derivational morphology, inflectional analysis, finite-state parsing, and Porter Stemming to group related words for semantic clustering.

Morphological Analysis

Word	Root	Affix	Type	Word class	Final stem
decide	decide	---	Base	Verb	decid
decision	decide	-ion	Derivational	Noun	decis
decisive	decide	-ive	Derivational	Adjective	decis
deciding	decide	-ing	Inflection	Verb	decid

1. Decide
decide
Base verb.
No added suffix.
Porter removes the final -e:
decide → decid
Final stem: decid

2. Decision
Morphological relationship:
decide → decision
-ion is a derivational suffix.
Forms a noun from the verb.
Porter processing produces:
decision → decis
Final stem: decis
.
3. Decisive
Morphological relationship:
decide + -ive → decisive
-ive is a derivational suffix.
Forms an adjective.
Porter processing:
decisive → decis
Final stem: decis

4. Deciding
decide + -ing → deciding
-ing is an inflectional suffix.
Represents the present participle/continuous form.
Porter processing:
deciding → decid
Final stem: decid

Finite-State Transitions
```
S0 → decide → Accept
       |
       ├── -ion → decision → Accept
       ├── -ive → decisive → Accept
       └── -ing → deciding → Accept
```

Accepted forms
↓
Porter Stemmer

↓
decid / decis

Final Output
decide → decid
decision → decis
decisive → decis
deciding → decid

Semantic Interpretation
decide → action of making a choice.
decision → noun representing the result/process of deciding.
decisive → adjective describing something characterized by decisiveness.
deciding → ongoing action of deciding.

The forms are semantically related, but the Porter Stemmer does not necessarily produce exactly the same stem for every derivational variant. It produces decid for decide/deciding and decis for decision/decisive. This distinction should be preserved rather than artificially forcing every form to the same root.

Rubrics for Calculation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Task	20%	Clearly understands the given task/experiment without assistance	Understands task with minor clarification	Partial understanding; requires guidance	Unable to understand the task
Procedure & Execution	25%	Performs procedure accurately and independently with proper technique	Performs with minor errors	Requires guidance; makes procedural mistakes	Unable to execute the procedure
Observation & Result	20%	Records accurate observations and obtains correct results	Minor errors in observation or result	Incomplete or partially correct results	Incorrect or no results
Viva Voce / Conceptual Response	20%	Answers all questions confidently with strong conceptual clarity	Answers most questions with minor gaps	Limited responses; needs prompting	Unable to answer questions
Time Management & Presentation	15%	Completes task on time with neat and clear presentation	Completes with minor delay or presentation issues	Delayed or poorly presented work	Unable to complete on time

Q. No. / Criteria Level	Understanding of Task	Procedure & Execution	Observation & Result	Viva Voce / Conceptual Response	Time Management & Presentation	Sub. Total	Total %
1							
2							
3							
4							
5							

Faculty In-charge	Course Coordinator	Program Director
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________